

Unload the Dishwasher to the Cabinet

SOP (1x Unload, in situ)

One episode empties one finished dishwasher where it is installed and puts every item away in the cabinet and the drawer beside it. The base is **passive**: it has no drive of its own, so it is pushed by hand up to the dishwasher and locked there, and nothing is carried away to a table. Everything the episode touches is already at the machine when recording starts: the dishwasher with its door open and both racks loaded, the empty cabinet on the counter to its right, and the open drawer below the cabinet with its empty cutlery insert.

The dishwasher is worked **as found**. The cycle has finished and the machine is off. The door already hangs open and flat when recording starts, both racks are pushed in, and the twelve clean, dry items sit in them as they were washed: two plates on edge in the plate slots, two bowls upside down in the bowl area, six utensils standing head down in the cutlery basket, and two mugs upside down on the upper rack.

This is an in-situ task, and three things follow from that. First, the machine is built in under a counter, so **nothing is taken out of a rack until that rack is drawn fully out past the counter edge**. No gripper reaches in under the counter, and no item is lifted out of a rack that is still under it. Second, **the machine, the cabinet, and the drawer are never leaned on and never pushed**: no gripper, wrist, or forearm rests on the door, a rack rail, the cabinet, or the drawer front, and no push is ever hard enough to move any of them. Third, **nothing is turned over**: a bowl or a mug that sits mouth down in the rack is put away mouth down. A plate or a utensil is laid flat by leveling the gripper that holds it, never by flipping the item over.

The order never changes: **plates, then bowls, then cutlery, then the lower rack goes in, then the upper rack comes out, then the mugs, then the upper rack goes in**. The upper rack is only drawn out after the lower one is pushed in, because an upper rack that is out sits over the lower one and blocks it.

Arm roles are fixed and never swap. **The right gripper does all the item work**: every plate, bowl, utensil, and mug out of the racks and into its place in the cabinet or the drawer. **The left gripper is the rack gripper**: it draws each rack out, holds it still, and pushes it back in. Neither gripper does the other's work.

Setup

Complete the base positioning and all the checklists before starting an episode.

Cell configuration

- **Environment camera**: 900 mm.
- **cell_type**: bimanual

Base positioning

The base is **passive**. It has no drive of its own, it is pushed into place by hand, and it is never steered, nudged, or repositioned once recording starts. It is parked once, before recording, and does not move again until the episode is over.

1. Push the base by hand up to the dishwasher and stop it **square to the front of the machine**, so the front edge of the open door runs straight across the frame and neither end of it sits nearer than the other.
2. Stop it **centered on the door mouth**, so the left end and the right end of the racks are the same distance out from the middle of the base.
3. Stop it **far enough back** that a rack drawn fully out swings clear of the base and of both arms, and the right side of the base stands clear of the open drawer.
4. Stop it **close enough** that the **right gripper** reaches the back of each rack when that rack is fully out, without the arm extending.
5. Check the **height band**: with the base parked, the **right gripper** comes down onto a fully out rack from above and reaches the floor of the lower rack, the cutlery basket, and the bed of the upper rack, all without the wrist or forearm fouling the counter edge above.
6. Check the **cabinet**: the **right gripper** slides level through the open front of the cabinet onto the plate spot on the lower shelf without touching the upper shelf, comes down onto the bowl spot on the upper shelf from above, and reaches both mug hooks, all without extending.
7. Check the **drawer**: the **right gripper** comes down onto the open drawer from above and reaches all three compartments of the insert without extending and without touching the drawer front.
8. Check the **rack rails**: the **left gripper** reaches both rack rails at their left ends and can draw each rack out to its stop and push it back in without extending.
9. Lock or brake the base. Push it firmly once by hand: it must not roll, creep, or turn.
10. If any of lines 1 to 8 fails, push the base to a new park by hand and start again at line 1. Do not work a station the arms cannot reach comfortably.

The base stays locked and still for the whole episode. Nothing moves it: no arm leans on the machine, the cabinet, the drawer, or the counter hard enough to shift it, nothing and nobody touches it, and it is never repositioned mid-task. A base that moves after recording starts ends the episode.

Hardware checklist

1. Cameras are on and recording.
2. The environment camera is centered on the door mouth and its frame includes the open door, both racks pushed in and fully out, the whole cabinet with both shelves and both hooks, and the open drawer with its insert.

3. The camera reads a fully out rack from above, so which slot a plate stands in, which way up a bowl or mug sits, and which compartment a utensil stands in are all readable.
4. The camera reads the inside of the cabinet, so the plate stack on the lower shelf, the bowl stack on the upper shelf, and a mug hanging from each hook are readable.
5. The camera reads the open drawer from above, so which compartment each utensil lies in is readable.
6. Both arms are at home with grippers open.
7. The **right arm** reaches the whole floor of the lower rack, the cutlery basket, the whole bed of the upper rack, the plate spot, the bowl spot, both hooks, and all three drawer compartments, without extending to a joint limit. It reaches neither a rack rail nor the door.
8. The **left arm** reaches both rack rails at their left ends without extending to a joint limit. It reaches nothing in the cabinet or the drawer.
9. The **right gripper** comes down onto a fully out rack straight from above, and never reaches in under the counter edge toward a rack that is still in.
10. The two arms do not collide, and a rack drawn fully out is clear of the right arm's path from the rack to the cabinet and to the drawer.
11. If a place cannot be reached, re-park the base by the Base positioning steps until lines 7 to 10 hold.

Materials checklist

1. The **dishwasher** stands where it is installed, built in under the counter. It is not moved, not leaned on, and not pushed at any point. The cycle has finished and the machine is off.
2. The **door** is hinged at the bottom. It hangs open and flat, inside face up, stays there on its own without being held, and stays open for the whole episode. Nothing in this SOP closes it.
3. Both racks are **pushed fully in**, and both roll out to a stop and back on their runners without being lifted.
4. A rack drawn fully out stands clear past the counter edge along its whole length, so the whole rack is unloaded from above with nothing overhead.
5. The **lower rack** is loaded:
 - two plates standing on edge in two neighbouring **plate slots**, both facing the front of the rack,
 - two bowls upside down in the **bowl area**, side by side and not nested,
 - six utensils standing head down in the **cutlery basket**, two per compartment: both forks in the fork compartment, both knives in the knife compartment, both spoons in the spoon compartment.
6. The **cutlery basket** is clipped to the front rail of the lower rack at its right end. Its three compartments are marked forks, knives, spoons.

7. The **upper rack** holds two mugs upside down over the tines, one on each half, clear of each other. Each mug has one handle, and the handles point any way.
8. Every item is clean, dry, and undamaged. No plate is chipped, no mug handle is cracked, and no tine is bent or missing. Nothing drips.
9. The **cabinet** stands on the counter immediately to the right of the machine, close to its right edge, with its open front facing the front. It has two shelves. The **upper shelf** is the top of the cabinet and has nothing above it. The **lower shelf** is the cabinet floor, under the upper shelf. Both shelves are empty, and the cabinet does not slide when a plate is slid onto a shelf.
10. Two **mug hooks** are fixed to the underside of the upper shelf, just behind its front edge, on the right half of the shelf. Their tips point up, both hooks are bare, and two mugs hung on them hang clear of each other and clear of the plate spot.
11. The **drawer** sits in the counter directly below the cabinet. It is drawn open before recording, far enough that the whole **drawer insert** stands past the counter edge with nothing overhead, and it does not roll on its own. Nothing in this SOP opens or closes it.
12. The drawer insert has three **compartments** in a row from left to right, marked forks, knives, spoons, all empty. The insert sits still in the drawer and does not lift when a utensil is laid in it.
13. The counter around the cabinet is clear, and the floor in front of the machine is clear and dry, so nothing that is dropped rolls under the base.

Workspace layout

- **Dishwasher:** built in under the counter, straight ahead of the parked base.
 - **Lower rack:** rolls out toward the arms, over the open door, to a stop past the counter edge. It has the **plate slots** across its back, the **bowl area** across its front left, and the **cutlery basket** clipped to the front rail at its right end.
 - **Upper rack:** rolls out over the lower rack, to a stop past the counter edge. It is one open bed of tines and holds the two mugs.
 - **Door:** hanging open and flat for the whole episode. The racks roll out over it.
- **Cabinet:** on the counter immediately right of the machine, open front facing the front, within reach of the right arm.
 - **Lower shelf:** the cabinet floor. Its left half is the **plate spot**, where the two plates are stacked flat.
 - **Upper shelf:** the top of the cabinet, nothing above it. Its middle is the **bowl spot**, where the two bowls are nested.
 - **Mug hooks:** two hooks under the front edge of the upper shelf, on its right half: the **left hook** and the **right hook**. A hung mug hangs down in front of the right half of the lower shelf.

- **Drawer:** in the counter directly below the cabinet, drawn open, within reach of the right arm. Its insert has the **fork compartment** on the left, the **knife compartment** in the middle, and the **spoon compartment** on the right.

The right arm works the racks in front of the machine, the cabinet, and the drawer. The left arm works the rack rails and stays on its own side of the machine.

Arm assignments

- **The right gripper owns the items.** It lifts every plate, bowl, utensil, and mug out of a rack, carries it, and puts it in its place in the cabinet or the drawer. It never touches a rack rail, the door, the drawer front, or the cabinet itself.
- **The left gripper owns the racks.** It draws each rack out, holds the rack still while the right gripper empties it, and pushes it back in. It never picks up an item and never reaches the cabinet or the drawer.
- **The left gripper holds the rack still the whole time that rack is being emptied.** It takes the rack rail at the rack's left end and keeps its hold until the rack is empty.
- An item is released only in its place: on the plate spot, on the bowl spot, hanging from its hook, or lying in its compartment. Never open a gripper over open space, over the open door, over a rack, or over the counter.

Vocabulary

- **In situ:** the machine is worked where it is installed. Nothing is taken out of the kitchen to a table, and nothing about the kitchen is rearranged for the task.
- **Passive base:** the base has no drive. It is pushed by hand to its park before recording, locked there, and it does not move at all during the episode.
- **Tines:** the rows of upright pegs standing in a rack.
- **Slot:** the gap between two neighbouring tines in the lower rack, wide enough for one plate standing on edge.
- **Plate slots:** the row of slots across the back of the lower rack, where the plates stand.
- **Bowl area:** the open stretch of tines across the front left of the lower rack, where the bowls sit upside down.
- **Rack rail:** the bar across the front of a rack. The left gripper takes the rail to roll the rack and to hold it still.
- **Fully out:** the rack has been drawn forward until it stops, and its whole length stands past the counter edge with nothing overhead.
- **Fully in:** the rack has been pushed back until it stops and its rail is level with the front of the cabinet of the machine.
- **Cutlery basket:** the open box clipped to the front rail of the lower rack. It has three compartments in a row, one per type, marked forks, knives, spoons.

- **Head down, handle up:** how every utensil stands in the basket: the eating end pointing down into the compartment, the handle standing above the rim.
- **Rim:** the outer edge of a plate. A plate on edge in a slot has its rim standing above the tines at the top, and that is where the right gripper pinches it.
- **Foot ring:** the raised ring on the underside of a bowl. A bowl upside down on the tines has its foot ring on top, and that is where the right gripper pinches it. **Unvalidated.**
- **Level the gripper:** turn the gripper from pointing down to pointing straight ahead, so an item hanging below it swings up to lie flat, pointing away from the gripper. The item is not flipped over and the gripper does not let go.
- **Flat:** a plate lying on its base, or a utensil lying on its side, with its whole length resting on the surface below it.
- **Plate spot:** the left half of the lower shelf, where the two plates are stacked flat.
- **Bowl spot:** the middle of the upper shelf, where the two bowls are nested.
- **Square:** the top plate of the stack lies centered on the plate below it, so no part of either rim sticks out past the other.
- **Nested:** the second bowl sits down over the first, upside down on it, so the two bowls make one stack and not two.
- **Mug hook:** a hook fixed under the front edge of the upper shelf with its tip pointing up. A mug hangs from it by its handle.
- **Hung:** the mug's handle is over the hook and resting on it, the gripper has let go, and the mug hangs still from the hook and touches nothing else.
- **Drawer insert:** the tray lying in the open drawer with three compartments in a row.
- **Compartment:** one of the three troughs of the drawer insert, or one of the three sections of the cutlery basket. Each is marked for one type: forks, knives, or spoons.
- **Seated:** the item is resting on the shelf, on the item below it, or on the floor of a compartment, and it stays still for 2 seconds after the right gripper opens. A seated item does not rock or lean on a neighbour.
- **Clear of:** not touching. Two items are clear of each other when a gap shows between them.

Handling standard

- Move exactly one item at a time.
- Nothing is turned over. A bowl or a mug goes to its place mouth down, as it sat in the rack. A plate or a utensil is laid flat by leveling the gripper, never by flipping it over.
- Unload only a rack that is **fully out**. No gripper reaches in under the counter, and nothing is lifted out of a rack that is still under it.
- Lift every item straight up until it is clear of the tines, carry it level and high enough to clear everything still in the rack, and lower it straight down into its place.

- Lower each item until it is resting, then release. Never drop an item onto a shelf, into a compartment, or onto a hook.
- Nothing is slid or dragged across tines, a shelf, or a compartment. If an item lands badly, the right gripper lifts it clear and lowers it again rather than nudging it into place.
- If the item due next leans on or is crossed by another, the right gripper lifts the other one straight up, stands it back down clear in the same place, and then takes the item that was due.
- Empty each rack fully before the left gripper pushes it in. Once a rack is in, nothing is taken from it.
- Nothing rests on the machine, the cabinet, or the drawer. No gripper, wrist, or forearm leans on the door, a rack rail, a shelf, or the drawer front, and no push is ever hard enough to move any of them.

Steps

Step 1: Draw the lower rack out and stack both plates on the lower shelf

Goal: the lower rack is fully out past the counter edge and both plates lie flat on the plate spot, one on the other.

1.1 Draw the lower rack out

- The **left gripper** takes the rack rail of the lower rack at its left end and draws the rack straight forward, over the open door, until it is **fully out**.
- Draw it level and in one movement. If it catches on its runners, push it back and draw it again. Never lift a rack to free it.
- The **left gripper** keeps its hold on the rail and holds the rack still.

Check: the whole rack stands past the counter edge with nothing overhead, and the rack is still. If any part of it is still under the counter, the **left gripper** draws it forward again.

1.2 Lay each plate flat on the plate spot

Work one plate at a time. Take the plate in the slot nearer the right end of the rack first.

- The **right gripper** comes down onto the rack from above and pinches the plate at the top of its **rim**, where the rim stands above the tines.
- Lift the plate straight up, still on edge, until its bottom edge is clear of the tines.
- Carry it on edge to the front of the cabinet, high enough to clear everything still in the rack.
- In front of the cabinet, **level the gripper** so the plate lies flat and points away from the gripper into the cabinet. The gripper keeps the same hold on the rim.
- Slide the plate in level through the open front onto the **plate spot**, keeping the plate and the gripper under the upper shelf without touching it.

- Lower the plate until it is resting, then release and draw the gripper straight back out of the cabinet.
- The second plate goes on top of the first, **square** on it.

Check: both plates lie flat on the plate spot, one on the other, square, and neither rim sticks out past the other. If the top plate is off, the **right gripper** pinches its near rim, lifts it just clear of the plate below, and lowers it again square. Never push a plate across the shelf.

Expected state: the lower rack is fully out with the **left gripper** still on its rail, the plate slots are empty, two plates lie stacked on the plate spot, and the bowls and the utensils are still in the rack.

Step 2: Nest both bowls on the upper shelf

Goal: both bowls sit upside down on the bowl spot, one nested on the other.

- The **left gripper** keeps its hold on the rack rail throughout.
- Work one bowl at a time. Take the bowl nearer the front of the rack first.
- The **right gripper** comes down from above and pinches the bowl by its **foot ring**, lifts it straight up until its rim is clear of the tines, and carries it level, still upside down, over the top of the cabinet to the **bowl spot**.
- Lower it straight down until it is resting, then release and lift the gripper straight up.
- The second bowl goes down over the first, so the two are **nested**, not side by side.

Check: both bowls are upside down on the bowl spot, nested, and seated. If a bowl perches on the edge of the other, rocks, or sits beside it, the **right gripper** pinches its foot ring, lifts it straight up, and lowers it again.

Expected state: the lower rack is still fully out, the plate slots and the bowl area are empty, two bowls sit nested on the upper shelf, and the six utensils are still in the cutlery basket.

Step 3: Lay the six utensils in the drawer insert by type

Goal: all six utensils lie flat in the drawer insert, two per compartment, heads toward the back of the drawer.

- The **left gripper** keeps its hold on the rack rail throughout.
- Work one utensil at a time, in **type order: both forks, then both knives, then both spoons**. Within a type, take the one nearer the right end of the basket first.
- If the utensil due next leans on or is crossed by another, the **right gripper** lifts the other one straight up, stands it back down clear in the same compartment of the basket, and then takes the one that was due.
- The **right gripper** comes down from above and pinches the utensil by its handle above the basket rim, lifts it straight up until its head is clear of the basket, and carries it upright to the drawer, over the **compartment** for its type.

- Over the compartment, **level the gripper** so the utensil lies flat with its head pointing to the back of the drawer and its handle toward the front.
- Lower it until it is resting on the floor of the compartment, then release and lift the gripper straight up.
- The second utensil of a type lies beside or on the first, inside the same compartment.
- Never drag a utensil sideways out of the basket, and never carry two at once.

Check: six utensils lie flat in the insert, two per compartment, heads to the back, forks in the fork compartment, knives in the knife compartment, spoons in the spoon compartment. No utensil lies across a divider or sticks out of the insert. If one is in the wrong compartment, the **right gripper** pinches its handle, lifts it straight up, and lays it in the right one.

Expected state: the lower rack is empty and still fully out, the drawer insert holds all six utensils, and both mugs are still on the upper rack.

Step 4: Push the lower rack in, draw the upper rack out, and hang both mugs

Goal: the lower rack is fully in, both mugs hang from their hooks, and the upper rack is fully in.

4.1 Push the lower rack in

- Confirm the lower rack is empty: no plate, bowl, or utensil is left in it.
- The **left gripper**, still on the rack rail, pushes the lower rack straight back until it is **fully in**, then releases and comes clear.
- Push it in level and in one movement. If it catches, draw it forward again, straighten it, and push it in again. Never force it.

4.2 Draw the upper rack out

- The **left gripper** takes the rack rail of the upper rack at its left end and draws it straight forward until it is **fully out**.
- The **left gripper** keeps its hold on the rail and holds the rack still.

Check: the whole upper rack stands past the counter edge with nothing overhead, and the lower rack is fully in beneath it.

4.3 Hang each mug on its hook

Work one mug at a time. Take the mug on the **right half** of the upper rack first; it goes on the **right hook**. The mug on the left half goes on the **left hook**.

- The **right gripper** comes down from above and grips the mug around its body, then lifts it straight up until its rim is clear of the tines.
- Turn the mug with the wrist so its handle points away from the gripper, toward the back. Keep the mug mouth down.

- Carry it level to the front of the cabinet, below the front edge of the upper shelf and in front of its hook.
- Raise the mug until the opening of its handle is higher than the tip of the hook, move it straight back until the hook is inside the handle, then lower it until the handle rests on the hook.
- Open the gripper and draw it straight back. Let the mug hang.
- Neither mug is set down on a shelf, and neither goes over a hook that already holds a mug.

Check: each mug is **hung** from its own hook by its handle and hangs still, clear of the other mug and of the plate stack. If a mug rests on the lower shelf, is caught by anything but its handle, or is on the wrong hook, the **right gripper** grips it around the body, lifts it off, and hangs it again.

4.4 Push the upper rack in

- Confirm the upper rack is empty.
- The **left gripper**, still on the rail, pushes the upper rack straight back until it is **fully in**, then releases and comes clear.

Expected state: both racks are fully in and empty, the door still hangs open and flat, the cabinet holds the plate stack, the bowl stack, and two hanging mugs, and the drawer insert holds six utensils.

Step 5: End the episode

Goal: the unload is complete, both racks are in, and both arms are safely home.

- Confirm both racks are fully in and empty, and the door still hangs open and flat.
- Confirm the cabinet holds two plates stacked square on the plate spot, two bowls nested on the bowl spot, and one mug hung from each hook.
- Confirm the drawer insert holds six utensils lying flat, two per compartment.
- Confirm the base has not moved: it is still square to the machine and still locked.
- Return both arms home, then stop recording.

After the episode: reset the workspace

This reset is not recorded. The base stays parked and locked through the reset, and is only pushed away by hand once the station is set for the next episode.

1. Draw the upper rack out. Lift both mugs off their hooks and stand them upside down on the upper rack, one on each half, clear of each other, handles pointing any way. Push the rack fully in.
2. Draw the lower rack out. Take the six utensils out of the drawer insert and stand them head down in the cutlery basket, two per compartment by type. Stand them so none leans on or

crosses another.

3. Take the two bowls off the upper shelf and set them upside down in the bowl area, side by side and not nested.
4. Take the two plates off the plate spot and stand them on edge in two neighbouring plate slots, both facing the front of the rack. Vary which two slots between episodes.
5. Push the lower rack fully in and leave the door hanging open and flat.
6. Check the cabinet shelves and both hooks are bare and the drawer insert is empty, and that the drawer still stands open with the whole insert past the counter edge.
7. Wipe the shelves, the insert, and the counter if anything is marked or wet.
8. Replace anything chipped, cracked, or bent, any rack with a bent or missing tine, and any hook that has worked loose.
9. Check both racks still roll out and in freely, the basket is clipped firmly to the front rail, and the cabinet has not shifted on the counter.
10. Check the base is still square, centered, and locked, and push it firmly once by hand to confirm it does not roll. If it has moved, park it again by the Base positioning steps.
11. Run the Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Base moved during the episode

- **Visible cue:** the base rolls, creeps, or turns after recording starts, the door mouth shifts in the frame, or an arm pushes off the machine, the cabinet, or the counter hard enough to move the base.
- **SOP rule broken:** Step 5 and the Base positioning rule it confirms, the base is parked and locked before recording and does not move at all during the episode.
- **Coaching note:** park it, lock it, push-test it. If the park is wrong, fix it before recording, never during.

Violation: Machine, cabinet, or drawer leaned on, pushed, or moved

- **Visible cue:** a gripper, wrist, or forearm rests on the door, a rack rail, a shelf, or the drawer front; an arm pushes off any of them; the drawer rolls in or out; or the cabinet or the open door visibly shifts under a push.
- **SOP rule broken:** Steps 1 to 4, nothing rests on the machine, the cabinet, or the drawer, and no push is ever hard enough to move any of them.
- **Coaching note:** nothing at the station holds an arm up. Take the weight on the arm.

Violation: Unload order broken

- **Visible cue:** the work does not run plates, then bowls, then cutlery, then lower rack in, then upper rack out, then mugs, then upper rack in. For example a bowl comes out before both plates are stacked, a utensil comes out before the bowls, or the upper rack is drawn before the lower rack is pushed in.
- **SOP rule broken:** Steps 1 to 4, the order is plates, bowls, cutlery, lower rack in, upper rack out, mugs, upper rack in.
- **Coaching note:** say the next stage out loud before you reach for anything. The order never changes.

Violation: Rack unloaded before it is fully out

- **Visible cue:** the right gripper takes an item out of a rack that is still in, or only part way out, or reaches in under the counter edge toward a rack still under it.
- **SOP rule broken:** Steps 1.1 and 4.2, the left gripper draws the rack fully out past the counter edge before anything is taken from it, and no gripper reaches in under the counter.
- **Coaching note:** rack all the way out into the open first, then come down onto it from above.

Violation: Rack not held still while it is emptied

- **Visible cue:** the left gripper lets go of the rack rail during Steps 1 to 3 or 4.3, or the rack rolls, shifts, or runs back in while the right gripper is lifting an item out.
- **SOP rule broken:** Steps 1 to 4, the left gripper holds the rack rail the whole time that rack is being emptied.
- **Coaching note:** the left hand stays on the rail until the rack is empty.

Violation: Rack pushed in with items still in it or left part way

- **Visible cue:** the lower rack is pushed in with a plate, bowl, or utensil still in it, the upper rack is pushed in with a mug still on it, something is taken from a rack after it was pushed in, or a rack is left standing part way at the end of its step.
- **SOP rule broken:** Steps 4.1 and 4.4, confirm the rack is empty, then push it fully in, and take nothing from a rack that is in.
- **Coaching note:** look in the rack, then all the way in, once.

Violation: Item put in the wrong place

- **Visible cue:** a plate goes on the upper shelf or on the right half of the lower shelf, a bowl goes on the lower shelf or in the drawer, a utensil goes anywhere but the drawer insert, or a mug is set on a shelf or hung anywhere but a mug hook.
- **SOP rule broken:** Steps 1.2, 2, 3, and 4.3, plates on the plate spot, bowls on the bowl spot, cutlery in the drawer insert, mugs on the hooks.
- **Coaching note:** plates low left, bowls on top, cutlery in the drawer, mugs on the hooks. Every time.

Violation: Plate not leveled and slid in from the front

- **Visible cue:** the plate is flipped over in the air instead of leveled, is pushed into the cabinet still on edge, is dropped onto the shelf from above the front edge, or the gripper or forearm scrapes the upper shelf on the way in.
- **SOP rule broken:** Step 1.2, level the gripper in front of the cabinet so the plate lies flat, then slide it in level under the upper shelf and lower it until it rests.
- **Coaching note:** flat first, then in level, then down. The plate never leaves the rim hold.

Violation: Plates not stacked square

- **Visible cue:** the second plate is set beside the first instead of on it, rests half off the plate below with a rim sticking out, leans against the cabinet side, or is pushed across the shelf to line it up.
- **SOP rule broken:** Step 1.2, the second plate goes on top of the first, square on it, and a plate that is off is lifted clear and lowered again, never pushed.
- **Coaching note:** one plate on the other, rims lined up. Lift to fix, never slide.

Violation: Bowl turned over or not nested

- **Visible cue:** a bowl is flipped mouth up on the way to the shelf, is set beside the first bowl instead of over it, or perches on the edge of the bowl below and rocks when the right gripper opens.
- **SOP rule broken:** Step 2, both bowls go to the bowl spot upside down, the second nested down over the first, and each is seated before release.
- **Coaching note:** the bowl is already upside down and stays that way. Straight down over the first one until it stops moving.

Violation: Cutlery taken out of type order

- **Visible cue:** the utensils leave the basket in an order other than both forks, then both knives, then both spoons.
- **SOP rule broken:** Step 3, work in type order, forks, then knives, then spoons, and within a type take the one nearer the right end of the basket first.

- **Coaching note:** forks, knives, spoons. Say the type before you reach.

Violation: Utensil in the wrong compartment

- **Visible cue:** a fork lies in the knife or spoon compartment, two types share one compartment, or one compartment is left empty while another holds three.
- **SOP rule broken:** Step 3, one type per compartment, two utensils in each of forks, knives, and spoons.
- **Coaching note:** read the compartment before you lower the utensil, not after.

Violation: Utensil not laid flat with its head to the back

- **Visible cue:** a utensil is released standing on its head in the compartment and falls over on its own, is laid with its handle to the back, lies across a divider, or hangs out over the front of the insert.
- **SOP rule broken:** Step 3, level the gripper over the compartment so the utensil lies flat, head to the back and handle to the front, and lower it until it rests before release.
- **Coaching note:** flat in the air first, then down. Never let go of a standing utensil.

Violation: Utensil dragged or pulled from under another

- **Visible cue:** a utensil is pulled sideways out of the basket, or tugged free from under one leaning on it, instead of the leaning utensil being lifted straight up and stood clear first.
- **SOP rule broken:** Step 3, lift the leaning utensil straight up, stand it back down clear in the same compartment, then take the one that was due.
- **Coaching note:** uncover it, then lift it straight up. Never tug.

Violation: Mugs hung in the wrong order or on the wrong hook

- **Visible cue:** the mug from the left half of the rack goes first, a mug is hung on the left hook while the right hook is still bare, or both mugs end up on one hook.
- **SOP rule broken:** Step 4.3, take the right-half mug first and hang it on the right hook, then the left-half mug on the left hook.
- **Coaching note:** far hook first, so the first mug is never in the way of the second.

Violation: Mug not hung by its handle

- **Visible cue:** a mug is turned mouth up on the way, is set down on the lower shelf, is caught on the hook by its rim or body instead of its handle, or the gripper opens before the handle is resting on the hook so the mug drops onto the hook or the shelf.
- **SOP rule broken:** Step 4.3, keep the mug mouth down, raise the handle opening above the hook tip, move back until the hook is inside the handle, lower until the handle rests on the hook, then release.
- **Coaching note:** up, back, down, then let go. The hook takes the handle, nothing else.

Violation: Item dropped in or dragged instead of lowered

- **Visible cue:** the right gripper opens above a shelf, a compartment, or a hook and the item falls the last stretch, or an item is slid or nudged across the tines, a shelf, or the insert into place instead of being lifted clear and lowered again.
- **SOP rule broken:** Steps 1 to 4, lower every item until it is resting, then release, and correct a bad placement by lifting it clear and lowering it again.
- **Coaching note:** all the way down, then open. Never nudge an item that is already down.

Violation: Item lifted before it is clear or carried through the rack

- **Visible cue:** a plate is swung sideways out of its slot before its bottom edge is clear of the tines, a bowl is dragged off the tines, a mug is pulled sideways through the tines, or an item is carried low across items still in the rack and knocks one.
- **SOP rule broken:** Steps 1 to 4, lift every item straight up until it is clear of the tines, then carry it high enough to clear everything still in the rack.
- **Coaching note:** straight up until it is free, then across. Plan the path before the lift.

Violation: Correction made without confirming the check

- **Visible cue:** a step's check is skipped: the episode moves on with a plate off square, a bowl rocking, a utensil across a divider, or a mug swinging on the wrong hook, and nothing is lifted and re-lowered.
- **SOP rule broken:** Steps 1.2, 2, 3, and 4.3, each check is confirmed before the next item is taken, and a bad placement is lifted clear and lowered again.
- **Coaching note:** look at it before you reach for the next one. Fix it now, not at the end.

Violation: Wrong arm used

- **Visible cue:** the left gripper picks up an item, or reaches into the cabinet or the drawer, or the right gripper takes a rack rail or the door.
- **SOP rule broken:** Steps 1 to 4, the right gripper handles the items, the left gripper handles the racks.
- **Coaching note:** right arm carries, left arm works the racks.

Violation: Item dropped, knocked over, or carried the wrong way

- **Visible cue:** an item falls out of a rack, off a shelf, off a hook, onto the open door, or onto the floor; the right gripper knocks a stacked plate, a nested bowl, or a hung mug while putting the next item in; or an item is carried over the open drawer or swung across the cabinet on the way.
- **SOP rule broken:** Steps 1 to 4, lift straight up, carry level to its own place, and lower straight down clear of what is already put away.

- **Coaching note:** carry high and level, come down straight, and stay clear of what is already in.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with an item still in a rack, a rack out or part way, a mug on a shelf, or an arm away from home.
- **SOP rule broken:** Step 5 (confirm both racks in and empty, the stacked plates, the nested bowls, the hung mugs, the six utensils in the insert, and the locked base; return both arms home; then stop recording).
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Base fault:** a brake that releases on its own, or a caster that seizes so the base cannot be parked square.
- **Defective object:** a chipped plate, a bowl with a broken foot ring, a mug with a cracked handle, a bent or missing tine, a basket that comes unclipped, a hook that pulls out of the shelf, or a drawer insert that lifts with the utensil. Replace or refix it before the next episode.
- **Machine or station fault:** a rack that jams on its runners, a drawer that rolls shut on its own, or a cabinet that slides on the counter under a correct level slide.

Annotation subtasks (from SOP)

1. Draw the lower rack fully out past the counter edge
2. Lift one plate out of its slot, level it, and lay it on the plate spot
3. Lift one bowl off the tines and nest it on the bowl spot
4. Lift one leaning utensil clear in the cutlery basket
5. Lift one utensil out of the basket, level it, and lay it in its compartment
6. Push the lower rack fully in
7. Draw the upper rack fully out past the counter edge
8. Lift one mug off the tines and hang it on its hook
9. Push the upper rack fully in
10. Return both arms home and end the episode